

Department of Electronic and Telecommunication Engineering
University of Moratuwa, Sri Lanka
EN3160 IMAGE PROCESSING AND MACHINE VISION
Course Outline—July 1, 2026

1 Introduction

Digital image processing helps us enhance images to make them visually pleasing, or accentuate regions or features of the image to better represent the content. For example, we may wish to enhance the brightness and contrast to make a better print of a photograph, similar to popular photo-processing software. In a magnetic resonance image of the brain, we may want to accentuate a certain region of image intensities to see the certain parts of the brain. Image analysis and computer vision, which go beyond image processing, help us to make decisions based on the contents of the image. Computer vision extensively uses machine learning, particularly deep learning. This course will first give you the basic knowledge required to enter into this exciting field, and equip you with basic tools to do image processing and computer vision. In the latter part of the course, we will learn to apply the knowledge in real problems.

This course is a pre-requisite for EN4584 Advances in Computer Vision. The follow-up module EN4554 Deep Learning for Vision will be offered in the semester 7. This will be helpful for EN3150 Pattern Recognition (co-offered) EN4574 Advanced Pattern Recognition offered in semester 7.

2 Outcomes

Aim: To equip the students with the basic knowledge and tools to process a digital image as done in common image processing software packages and implement simple machine vision algorithms.

After completing this course you will be able to

- Apply image processing algorithms for image enhancement,
- Apply classic vision techniques in feature detection and applications,
- Apply machine vision algorithms for classification and object detection, and
- Design machine vision solutions for common industry problems.

3 Contents

1. Early vision
 - (a) Basics of digital images and image processing
 - (b) Sampling, interpolation
 - (c) Point operations, linear filtering (e.g., convolution)
 - (d) Detection of edges, corners, blobs
 - (e) Cameras, light, and color
 - (f) Feature extraction
 - (g) Optical flow
 - (h) Morphological processing
 - (i) Frequency domain processing
 - (j) Introduction to image compression
2. Fitting and alignment
 - (a) Fitting: least squares, total squares, RANSAC
 - (b) Voting methods for fitting: Hough lines
 - (c) Alignment, transforms
 - (d) Image stitching
3. Multiple-view Geometry
 - (a) Camera models, calibration
 - (b) Epipolar geometry
 - (c) Two-view stereo
 - (d) Structure from motion
 - (e) Dense 3D reconstruction (neural radiance fields, Gaussian splatting)
4. Segmentation

- (a) Thresholding
 - (b) Region growing
 - (c) Active contours
 - (d) Introduction to level-set and graph-cut based segmentation
 - (e) Deep learning based segmentation
5. Introduction to deep learning for vision
- (a) Bag of visual words
 - (b) Linear classifiers
 - (c) Convolutional neural networks for classification
6. Introduction to deep learning for vision
- (a) Two-stage object detectors (R-CNN, Fast R-CNN, Faster R-CNN)
 - (b) Single-stage detector (YOLO, SSD)
7. Recent topics in computer vision (time permitting)
- (a) Vision transformers
 - (b) CLIP, SAM, VLMs
 - (c) Generative models (diffusion).

4 Prerequisites

Mathematics. Programming skills in Python.

5 Contact Hours, Course Material, Etc.

Instructor:	Ranga Rodrigo. ranga@uom.lk, 011 264 0422.
Teaching assistant:	Sandun Herath. herathhmsi.21@uom.lk Hasitha Gallella. gallellammhbb.21@uom.lk
Lectures:	2 hours per week: Wednesday 1:15 p.m. to 3:15 p.m.
Tutorials:	2 hours per week (lectures will take place on some days) Thursday 1:15 p.m. to 3:15 p.m.
Labs:	Help sessions for assessments on pre-arranged time slots and on your own for assignments and the project.
Website:	https://online.uom.lk/course/view.php?id=33200

6 Assessment Criterion

Continuous assessments: 40%, final closed-book written examination: 60%.

Continuous Assessments

Item	Date	Weight	Comments
In-class quizzes	Surprise	20%	Easy, e.g., best 80%
Assignment A01: Intensity Transformations and Neighborhood Filtering	July 24	5%	Easy
Assignment A02: Fitting and Alignment (RANSAC, homographic transformation, image stitching)	Aug. 7	5%	Easy
Assignment A03: Deep Learning for Computer Vision	Aug. 28	10%	Easy
Project mid-review (in pairs)	July 28	10%	Moderately difficult
Tutorials	Throughout	20%	Easy
Project final presentation and viva (in pairs)	Sept. 11	30%	Moderately difficult

Notes

Project topics will be given.

Paper option: if a student pair writes a good-quality paper and convinces the instructors of the novelty, full marks for the continuous assessments will be given.

7 Books

I will base this course on Prof. Svetlana Lazebnik's course on computer vision (<https://slazebni.cs.illinois.edu/fall24/> Fall 2024). Some slides will also be from this course or its earlier versions. Torralba et al. [2024], Gonzalez and Woods [2008], Forsyth and Ponce [2003], Szeliski [2010], Jain [1986], Sonka et al. [2007], and Goodfellow et al. [2016].

Bibliography

David A. Forsyth and Jean Ponce. *Computer Vision: A Modern Approach*. Prentice Hall, 2003. ISBN 0-12-085198-1.

Rafael C. Gonzalez and Richard E. Woods. *Digital Image Processing*. Prentice-Hall Inc., Upper Saddle River, New Jersey 07458, 3rd edition, 2008. ISBN 978-81-203-3640-7.

Ian Goodfellow, Yoshua Bengio, and Aaron Courville. *Deep Learning*. MIT Press, 2016.

Anil K. Jain. *Fundamentals of Digital Image Processing*. Prentice-Hall Inc., Englewood Cliffs, New Jersey 07632, 1986. ISBN 0-13-336165-9.

Milan Sonka, Vaclav Hlavac, and Roger Boyle. *Image Processing: Analysis and Machine Vision*. HB, 3rd edition, 2007. ISBN ISBN/ISSN 0-495-08252-X.

Richard Szeliski. *Computer Vision: Algorithms and Applications*. Springer, first edition, 2010.

Antonio Torralba, Phillip Isola, and William T. Freeman. *Foundations of Computer Vision*. CMIT Press, 2024. ISBN 978-0-262-04897-2.